

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. What is the area, in square centimeters, of the triangle?

Answer

- A. **700**
- B. **350**
- C. **175**
- D. **80**

Correct Answer: B

Rationale

Choice B is correct. The area, ***A***, of a triangle is given by **$A = \left(\frac{1}{2}\right)bh$** , where ***b*** is the length of a base of the triangle and ***h*** is the corresponding height of the triangle. It's given that a triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. Substituting **10** for ***b*** and **70** for ***h*** in the formula **$A = \left(\frac{1}{2}\right)bh$** yields **$A = \left(\frac{1}{2}\right)(10)(70)$** , or **$A = 350$** . Therefore, the area, in square centimeters, of the triangle is **350**.

Choice A is incorrect. This is the product of the given base and height of the triangle, not its area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the given base and height of the triangle, not its area.